Nicolas Huber

IT Trainee | Data Engineering & ML Systems

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Profile

IT trainee focused on data engineering: data pipelines and infrastructure for ML systems (dbt, Dagster, Python, Kubernetes/OpenShift). MSc UZH summa cum laude; sped up model training by a factor of 12 in the Migros traineeship.

Work Experience

IT Trainee, Migros-Genossenschafts-Bund, Zurich 06/2025 – present

- Rotation programme with three assignments in data engineering and data-driven products

Data Team, Digitec Galaxus 07/2026 – present

- Migrated and modernised two ML classifiers (Python, LLM integration) to a new Airflow platform
- Found the cause of daily downgrades of in-demand products through SQL analysis on BigQuery; fix scheduled as a follow-up ticket
- Built automatic detection of purchase price errors from partner data (Airflow, BigQuery, Streamlit); errors are reported daily
- Method for identifying problematic stock items for the purchasing strategy (ongoing)

Data Engineering, Migros Bank, Product & Marketing Analytics 01/2026 – 06/2026

- Technical contact for building the data infrastructure of a team of about 10 analysts
- Built the team's first data pipeline: modelling with dbt, orchestration with Dagster on a new OpenShift platform
- Deployed a campaign tool on OpenShift to automate standard campaigns
- Exploratory analysis for customer segmentation and preparation of an ML model for validation in an A/B test
- Led workshops on dbt, data products and Git; the teams now work with Git and dbt

Data Engineering, MGB, Data & Analytics Division 06/2025 – 12/2025

- Introduced an aggregation layer for a promotion optimisation tool: cut the number of trained models from 2.4 million to 10,000
- Reduced the runtime of a training run from 3 h to 15 min (factor 12); implemented as a dbt pipeline on Google Cloud

Research Assistant (part-time), Interactive Visual Data Analysis Group, UZH 10/2023 – 12/2024

- Prototypes for visual analytics with ML and LLM components (Python, FastAPI, scikit-learn, Qdrant, Docker, Vue, D3)

Junior Web Developer (part-time), novu, Zurich 06/2021 – 01/2024

- Maintained and extended the internal ERP system (Symfony, PHP, JavaScript)
- Taught courses for application development apprentices at the ICT-Berufsbildungscenter

Education

MSc Software Systems & Data Science, University of Zurich 09/2022 – 04/2025

- Grade 5.7/6, summa cum laude (top 10 % of the cohort)
- Master's thesis (6.0/6): gamified system for AI-assisted SDG classification (FastAPI, Vue, Qdrant, LLM)
- Master's project (5.75/6): supply management system for health workers in rural Lesotho (ComBaCaL); built most of the admin panel and API in a team of five (Symfony, API Platform, MariaDB, CouchDB)
- Exchange semester at the University of Technology Sydney (02–06/2024), High Distinction in all subjects

BSc Software Systems & Information Systems, University of Zurich 09/2019 – 09/2022

- Grade 5.0/6, magna cum laude
- Bachelor's thesis (5.5/6) on Moving Target Defense for IoT devices; basis of an IEEE publication (2023)

Skills

Data	dbt, SQL, BigQuery, Airflow, Dagster, Python (scikit-learn), feature engineering
Platform	OpenShift/Kubernetes, Docker, Google Cloud, Git
Other	Streamlit, FastAPI, Vue, D3, Qdrant, Symfony/PHP
Languages	German (native), English (C1, Cambridge Advanced)